Selected Topics in Reinforcement Learning - Final Project

Introduction

Get ready for a thrilling project where you'll be training a model to take the wheel in a race car game! This is a first-person perspective racing game, where each step offers you only the current view. There are no boundaries - you can choose any algorithm or training framework. So, put on your thinking cap, strive for the top scores, and outshine the competition. Let's hit the track!

Lab Description

- Given the environment modified from ***racecar_gym***, train a neural network to control the racecar.
- Race against the clock and cover the longest distance you can!
- Observe environmental characteristics and choose your method.
- Design your mechanism to train a neural network to control the racecar.
- Try to apply different potential methods!

Important Date

- Available demo time
  - 12/11, 12/18, 12/25 (Mon.) evening
  - 2024/1/2 (Tue.) class time
  - You need to fill out the form in advance, just like the previous labs)
- Report
  - Submission deadline: 2024/1/5 23:59

Turn in

- You have to turn in your report (.pdf)
- Notice: zip all files with the name “RL_Project_StudentId_Name.zip”
  - E.g., “RL_Project_311551043_廖唯辰.zip”

    “RL_Project_311551043_廖唯辰.zip”

    |_____ report.pdf
  - Wrong format deduction: -5pts

Scoring Criteria

Your total project score = report * 0.4 + demo performance * 0.6

The focus of the demo is on performance, and the report part will pay more attention to the effort spent and things learned.

Part 1: Performance (demo)

- You **NEED to DEMO** to get performance grades
- Different maps can be tested with **different** models
- **Pre-recorded action sequences are prohibited** during the demonstration
- You have several chances to demo, and the **highest** score will be chosen in the end. (including the bonus)
- The distribution of demo scores
  - `Circle_cw` map:
    - Pass the Circle_cw threshold: **20%** of the demo score
    - Rank: **30%** of the demo score
  - `Austria` map:
    - Pass the threshold 1/2: **15%** of the demo score
    - Pass the threshold 2/2: **5%** of the demo score
    - Ranking: **30%** of the demo score
  - Bonus:
    - If you use the same model to evaluate both maps and you pass **both the Circle_cw threshold and Austria threshold 1/2**: **+5%** of the demo score

Part 2: Report

- In this section, we mainly score based on the effort and learning people put into "**Reinforcement Learning**" and "**Neural Network Training**".
  - If your content DOES NOT relate to at least one of the topics mentioned above, you will receive NO score in the report part. For instance, you might craft an "if-else" program that could potentially achieve

a decent score in the demo performance. However, if that is your only effort, you will receive **no points** in the report section.

- Report content and score distribution
  - Methodology Introduction (**20%** of the report score)
    - Introduction of the methodology used for each map
    - Explanation for the choice of methods for each map
  - Experiment Design and Implementation (**20%** of the report score)
    - Description of the training processes
    - Neural network architectures
    - Details of the hyper-parameters
    - List of packages, tools, or resources used
  - Method Comparison and Evaluation (**35%** of the report score)
    - Trying different methods and showing their results (It's better to include a diagram.)
    - Comparing the effectiveness of different approaches
    - Analyzing the successful and unsuccessful cases
    - Discussing the key observations and insights
  - Challenges and Learning Points: (**15%** of the report score)
    - Encountered challenges during the implementation process
    - The learnings from these challenges
  - Future Work: (**10%** of the report score)
    - The proposal of ideas for potential improvements
    - The suggestion of additional research directions for the future

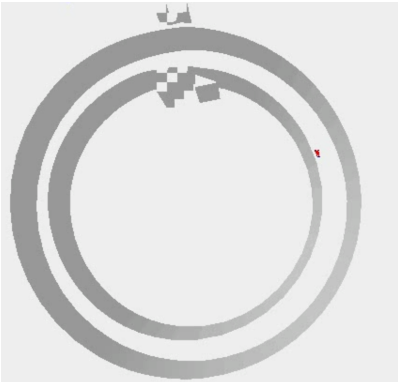
Environment

- The environment is modified from racecar_gym.
- When scoring, a pre-set parameter environment is used (see demo for details).
- Evaluation settings
  - Observation: a single 3x128x128 image
  - Action: `motor` and `steering` (both range from -1 to 1).
  - Default Reward: progress reward (difference of progress)
    - You may adjust the reward function according to your needs during your training sessions.
    - During the evaluation, we look at the progress you have completed when the time ends, not the total reward.

    - Terminated:
        - Maps:
            - `Circle_cw`: env time limit, student time limit, **collision**
            - `Austria`: env time limit, student time limit
        - Please refer to the demo part
    - Info: velocity, acceleration, pose, time, progress, lap, …
    - 0.02 sec a frame
    - During evaluation, we **may introduce some randomness**, such as adding a bit of noise to your action or randomizing the starting point. Therefore, please ensure to train a stable model!
        - **The randomness used may vary on different demo days.**
- For more information, you can read the code or the official GitHub page.
    - Note that the scoring is done **with the environment we provide** to everyone. The original GitHub is for reference only.
    - For training, you can change the environment to help you train. But be aware that the environment for scoring is preset.
- When you're training or testing, please set `render_mode = "rgb_array_birds_eye"`
    - Do NOT use render_mode = "human" in training or testing

Map: Circle_cw

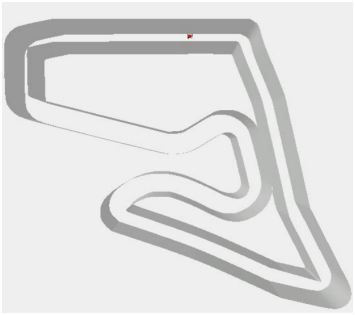
- Threshold: finish at least 1 lap

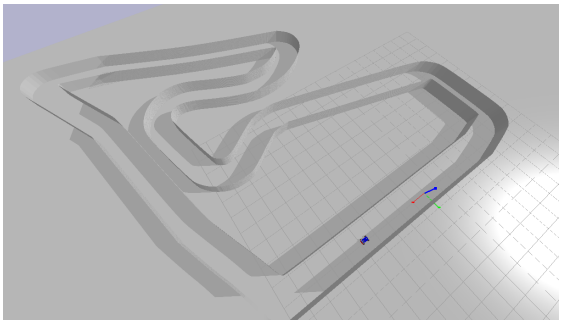


Map: Austria

- Threshold 1/2: pass the first curve (no collision)
- Threshold 2/2: pass the second curve (no collision)

The curve on the right side of the red car in the image below is the first curve; the lower right corner of the image below is the second curve.





The starting point is this car's position

Demo

The basic process during the demo

- Connect to the server opened by the TA (refer to server.py and client.py)
  - The *server* will return an `observation` to the *client*
  - The *client* sends an `action` to the *server*, then the server responds with a `terminated`

  ※ Please refer to the part about how to use server.py and client.py
- Explain the program and your design to the TA
- For the Circle_cw and Austria environments, **different models can be used** to score.
  - As mentioned in the previous part of the scoring criteria

Evaluation settings

- Env time limit: The time that the racing car can run in the game; that is, the time in the environment.
  - Note that every 0.02 seconds a frame
- Model time limit: the real-world time spent by the client
  - Timing Principle: Accumulate the time **between the server providing an observation and the reception of the subsequent action**
  - Your network delay may affect this
  - You may need to check the network settings to connect to our server.

  ※ Model Time Limit may be adjusted depending on the real cases
- Score: Lap + progress - 1 at the end of time
  - e.g., if you ran halfway (progress = 0.5) in the first lap and the time is up, then your score is 1 + 0.5 -1 (because the lap starts counting from 1)

The detailed settings:

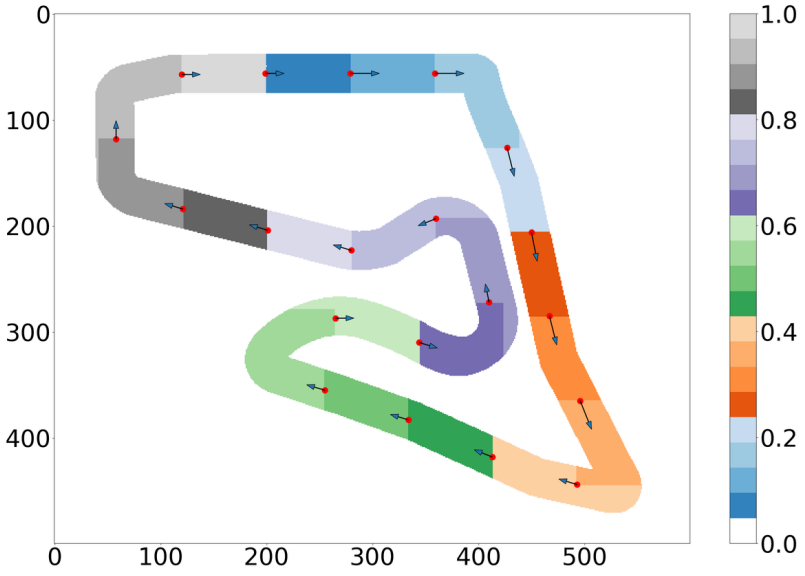
Map	Env Time Limit	Model Time Limit (min)	Recover	CollisionStopWhenEval
Circle_cw	25 sec	10 min	No	Yes

Map	Env Time Limit	Model Time Limit (min)	Recover	CollisionStopWhenEval
Austria	100 sec	20 min	Yes	No

Name	Scenario	reset_when_collision
Circle_cw	circle_cw_competition_collisionStop	False
Austria	austria_competition	True

About the collision in Austria

- In Austria, it allows wall collisions, but there will be penalties. The harder you hit, the more seconds will be deducted from your remaining time.
  - Velocity: (x, y, z, roll, pitch, yaw)
  - Time Penalty $= 30 + 10 \times \sum_{v \in \text{velocity}} v^2$
  - e.g., In Austria, hit the wall at the 20th second with velocity = [2,0,0,0,0,0], the remaining seconds to run are: $100 - 20 - (30 + 10 \times 4) = 10$ sec
- This picture of Austria is divided into 20 parts according to progress. In Austria, after a collision, the car will be revived to the next area.



Austria's 20 sections of track

Leaderboard

- After the start of the demo, the weekly updated ranking information will be announced.

RL-Topics Final Project Rank

Rank	SID	Score	Env Time	Acc Time	Video
1	0716092	0.6507	39.580	239	Link

Basic Usage

Install

```
pip install -e .
```

- Suggested environment:
  - Ubuntu 22.04
  - Python 3.10
  - gymnasium==0.28.1
- You can refer to the `Dockerfile` to construct your container.
  - You can refer to `container.md`

Construct the gymnasium env: `RaceEnv`

```
# Ensure you have installed racecar_gym so you can directly import the env
from racecar_gym.env import RaceEnv

env = RaceEnv(
    scenario=SCENARIO, # e.g., 'austria_competition', 'circle_cw_competition_collisionStop'
    render_mode='rgb_array_birds_eye',
    reset_when_collision=True, # Only work for 'austria_competition' and 'austria_competition_collisionStop'
)
```

Then, you can interact with it, for example:

```
obs, info = env.reset()
terminated = False
while not terminated:
    action = (1.0, 0.0) # Motor and steering
    obs, rew, terminated, truncated, info = env.step(action)
```

Evaluation by Yourself

- `server.py`
  - Open the environment and address client queries.
    - Provide observations if requested by the client.

        - Receive actions and respond with termination information.
    - Arguments:
        - `--port`: port
        - `--sid`: student id or any description
        - `--scenario`: what scenario to run
    - If you want to test for several episodes, please close the program and **restart it after each test**.
    - This is written using the Flask framework.
    - In the demo stage, the TA will open the server for you.
- `client.py`
    - Argument: `--url`: what address to connect
    - The purpose of client.py is to interact with the server using **HTTP** requests, transmitting action, and getting observations.
    - You can write your client, as long as you can correctly interact with the server.
    - An example (random agent) is given default.
        - You have to **modify it to use your neural network model**.
- Example of evaluating the random agent
    1. Run the server

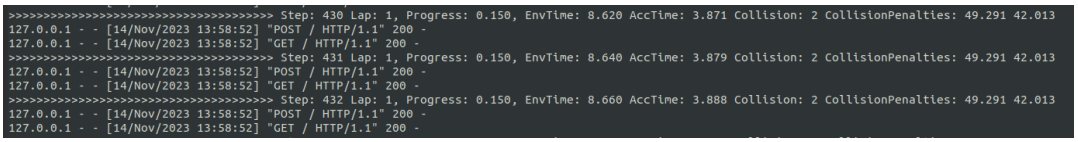
        ```
        # Example
        python server.py --port 33333 --sid tester--scenario austria_competition
        ```

    2. Run the client

        ```
        python client.py --url http://127.0.0.1:33333
        ```

    3. If they interact successfully:

        You can view information about the current status, as shown below, in the server's output:

    4. When they are running, you can go to `${url}/realtime` to watch the current testing observation.

        Example: open the web browser and go to `http://127.0.0.1:33333/realtime`

Evaluation Observation

5. After the episode is terminated, the video will be saved.

```
>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>> Step: 434 Lap: 1, Progress: 0.150, Envt
============ Terminal ============
Video saved to results/Tester_20231114-135852_env8.700_acc4s_score0.1498.mp4!
==================================
```

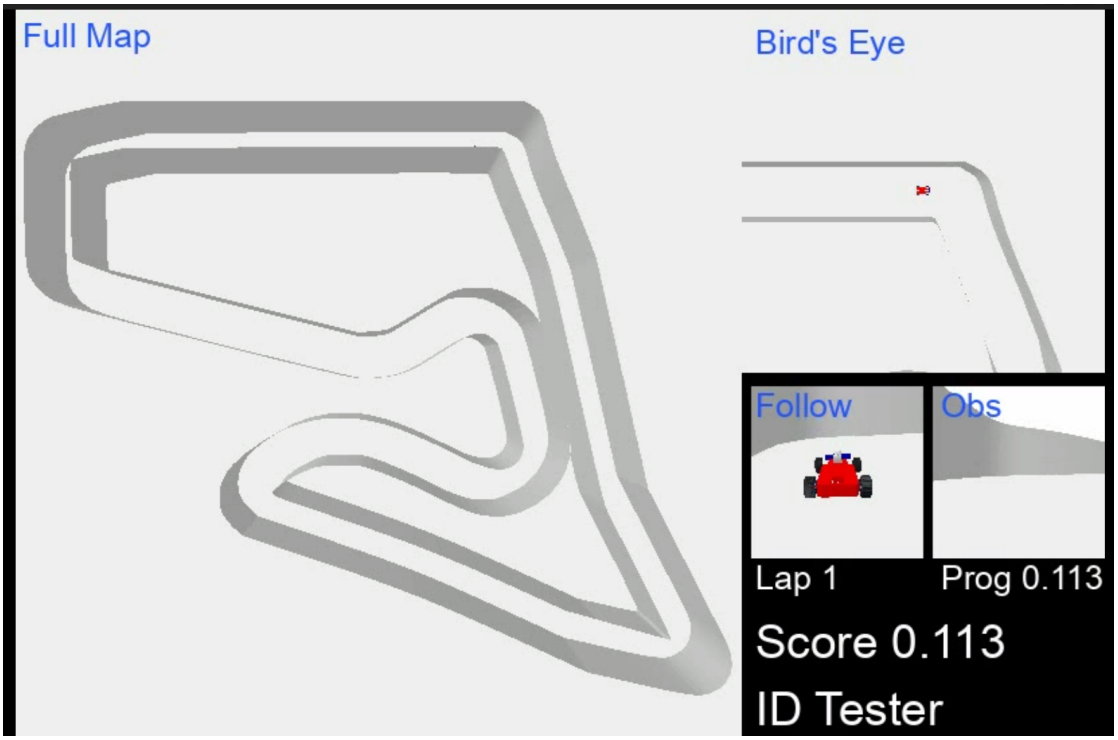
server

The client's output will display "Episode finished.”

```
Episode finished.
```

6. Play the video and you can see the episode

   The video contains multiple perspectives, the score, and the ID.



Note

- Not limited to algorithms.

- Not limited to programming frameworks, you can carve your own RL algorithm, or use other packages (such as Ray RLlib, Stable-Baselines3, and so on)
- This is the first time this project has been given, if there are any questions, feel free to respond
- Depending on the situation, rules may be adjusted or more information may be provided to everyone
- Feel free to ask questions in the discussion area!

Reference

- A. (n.d.). GitHub - axelbr/racecar_gym: A gym environment for a miniature racecar using the pybullet physics engine. GitHub. https://github.com/axelbr/racecar_gym

※ The cover photo is by toine G on Unsplash